

Comprehensive Learning Plan

I'll organize all these topics into a structured learning plan, grouping related concepts and removing duplicates.

■ JavaScript (130+ topics)

Core Concepts

- Primitive & Non-primitive data types
- Pass by value vs reference
- var, let, const differences
- Scope (block, function, global)
- Shadowing and illegal shadowing
- undefined vs not defined vs null
- TDZ (Temporal Dead Zone)
- Execution context
- Hoisting
- Lexical environment
- Closure (advantages, drawbacks, applications)
- this keyword (context, substitution)
- GlobalThis vs this
- Strict mode benefits

Functions

- Function types (declaration, expression, arrow, anonymous, named)
- First-class functions
- Higher-order functions
- IIFE (benefits)
- Factory functions
- Constructor functions
- Generator functions
- Currying (advantages)
- Memoization
- Callback functions
- Callback hell and solutions
- Pure functions

Objects & Arrays

- Object methods (keys, values, entries, freeze, seal)

- Object.create vs Object.assign
- Prototype and prototypal inheritance
- Prototype chain
- __proto__ vs prototype
- Array methods (map, filter, reduce, some, every, find, flat, flatMap)
- Shallow vs deep copy
- StructuredClone
- Object destructuring
- Optional chaining
- Nullish operator

Async Programming

- Event loop
- Call stack
- Microtask vs macrotask queues
- setTimeout, setInterval, setImmediate
- Promises (states, methods: all, race, any, allSettled)
- Async/await
- Error handling (try/catch/finally)
- Promise chaining

Advanced Concepts

- Proxy object and traps
- Reflect API
- WeakMap and WeakSet
- Symbol data type
- Iterators and generators
- Tagged templates
- Unicode handling
- Coercion (implicit/explicit)
- Truthy/falsy values
- NaN properties
- Double exclamation (!!)
- Lambda expressions
- Unary functions

DOM & Browser APIs

- DOM manipulation
- Event propagation (bubbling, capturing)

- Event delegation
- Event listener
- BOM (Browser Object Model)
- Web APIs
- Web Workers
- Service Workers
- Geolocation API
- Web Storage (localStorage, sessionStorage)
- IndexedDB
- Web Sockets

Practical JS Problems

- Find second largest/smallest in array
- Palindrome check
- String reversal ("Hello world" to "world Hello")
- Array flattening without flat()
- Remove duplicates without Set
- Remove consecutive odd numbers
- Check empty object
- Find max mark from object array
- Count character frequency
- Longest word in sentence
- Anagram check
- Fibonacci sequence
- Prime number check
- Random OTP generation
- Date manipulation (25 days ago)
- Decimal point check
- URL encoding
- Remove vowels from string
- Capitalize first letters
- Balanced parentheses check
- Find first non-repeating character
- Deep copy implementation
- Currying examples
- Debouncing implementation
- Throttling implementation
- Promise timing examples

■ Node.js & Express (100+ topics)

Core Node.js

- Node architecture
- V8 engine
- REPL
- Single-threaded nature
- Concurrency model
- LIBUV library
- Event-driven architecture
- Event emitter
- Process object
- Thread pool
- Worker threads
- Clustering
- Child processes (fork, spawn, exec, execFile)
- Buffer class
- Streams (types, piping, chaining)
- Back pressure
- Zlib compression
- Crypto module
- Net module
- OS module
- Util module
- Process.nextTick
- Error-first callbacks

Modules & NPM

- CommonJS vs ES modules
- Core modules
- require vs import
- package.json vs package-lock.json
- Dependencies vs devDependencies
- NPM vs Yarn
- Nodemon
- PM2 process manager

Express.js

- Express features
- Routing

- Middleware (types, application, router-level, error handling)
- app.use vs app.set vs app.all
- app.locals
- express.static
- express.json vs express.urlencoded
- Body parser
- Cookie parser
- Multer for file uploads
- Route parameters
- Query parameters
- Dynamic routing
- Router chaining
- View engines (EJS, HBS)
- Session management
- Authentication vs authorization
- CORS (preflight requests)
- CSRF protection
- Rate limiting
- Content negotiation
- HTTP methods (GET, POST, PUT, PATCH, DELETE, OPTIONS, HEAD)
- PUT vs PATCH difference
- HTTP request/response structure
- Status codes (200, 201, 400, 401, 403, 500)
- REST principles
- Web API vs REST API
- JWT (structure, signature, claims)
- HTTPS/TLS
- Environment variables
- Morgan logger
- Debug module

Practical Node/Express

- Create HTTP server
- Create Express server
- File operations (read/write sync/async)
- Logging middleware
- Authentication middleware
- Error handling middleware
- Multiplication table endpoint
- API to add numbers

- Process.nextTick example
- Stream piping example
- Current date to file
- Route protection with custom headers
- Reactor pattern implementation
- Control flow functions
- Cron jobs
- Promisify examples

■ MongoDB (70+ topics)

Core Concepts

- NoSQL vs SQL
- CAP theorem
- Partition tolerance
- ACID properties
- Isolation levels
- Namespace concept
- Document structure
- BSON format
- Max document size

Operations

- CRUD operations
- Update operators (\$set, \$inc, \$push, \$pull, \$pop)
- UpdateOne vs UpdateMany
- Upsert operation
- BulkWrite operations
- find() vs findOne()
- Projection
- Aggregation framework
- Aggregation pipeline stages
- \$match, \$group, \$project, \$lookup, \$unwind
- \$facet for multiple aggregations
- Single-purpose aggregation (count, distinct)
- \$expr for aggregation expressions
- \$addFields stage
- \$out stage

Indexing & Performance

- Indexing benefits
- Index types (hashed, geospatial, clustered)
- Covered queries
- explain() method
- Query optimization
- Drawbacks of indexing
- Journaling
- Write concern
- Read concern

Data Modeling

- Embedded vs referenced documents
- Normalization drawbacks
- GridFS for large files
- Capped collections
- isCapped method
- TTL indexes
- Transaction support

Practical MongoDB

- Find second highest salary
- Average price by genre
- 10% salary increment query
- Date queries (15 days before)
- Developers who know Java or Python but not both
- Developers joined after 2023
- Reduce salary by 30% for non-fullstack
- Rename collection
- mongodump usage
- \$lookup examples
- \$unwind examples
- \$addToSet examples
- Distinct values query
- Group by operations
- Create TTL index

■ React.js (120+ topics)

Core Concepts

- Virtual DOM
- Reconciliation process
- JSX syntax
- Components (functional vs class)
- Props and prop drilling
- State and immutability
- Event handling (synthetic events, native events)
- Event pooling
- Keys and why index shouldn't be used as key
- Refs (usage, forwardRef)
- Lifecycle methods (class components)
- Hooks rules
- Context API
- Error boundaries
- PropTypes
- React Fiber architecture

Hooks

- useState
- useEffect (cleanup, dependencies)
- useContext
- useReducer
- useCallback
- useMemo
- useRef
- useImperativeHandle
- useEffect
- Custom hooks

Routing

- React Router DOM
- MemoryRouter
- Route parameters
- Query parameters
- Navigation
- Conditional routing

State Management

- Redux principles

- Store, actions, reducers
- connect() function
- mapStateToProps, mapDispatchToProps
- useSelector, useDispatch
- Middleware (redux-thunk, redux-saga)
- Redux Toolkit (configureStore, createSlice, createAsyncThunk)
- React Context vs Redux
- LocalStorage vs Redux

Performance

- React.memo
- PureComponent
- useMemo for optimization
- useCallback for function caching
- Code splitting
- Lazy loading
- React optimization techniques

Advanced Concepts

- Higher-Order Components (HOCs)
- Render props pattern
- Portals
- Profiler API
- Concurrent features
- Automatic batching
- Transpiler vs compiler (Babel)
- Webpack basics
- SEO challenges and solutions
- Shadow DOM vs Virtual DOM
- HTML sanitization
- Internationalization

Practical React

- Reading query parameters
- Component communication (parent-child, siblings)
- Toggle custom hook
- Form handling
- API calls with axios (interceptors, cancel tokens)
- Authentication implementation

- Debounced search input
- Infinite scrolling
- Modal implementation
- Dynamic forms
- Drag and drop
- Chart integration
- Context for theme switching
- Error boundary implementation
- Custom hook for API calls
- Performance monitoring

■ Data Structures & Algorithms (150+ topics)

Complexity Analysis

- Big O notation
- Time complexity analysis
- Space complexity analysis
- Asymptotic analysis
- $O(1)$, $O(n)$, $O(\log n)$, $O(n \log n)$, $O(n^2)$, $O(2^n)$
- Complexity comparison

Memory Management

- Static vs dynamic memory allocation
- Stack vs heap memory
- Memory leaks
- Garbage collection (mark and sweep)
- Circular references
- Units of memory (byte, kilobyte, etc.)

Arrays

- Static vs dynamic arrays
- Operations complexity
- Multi-dimensional arrays
- Jagged arrays
- Sparse arrays
- Typed arrays
- Array manipulation algorithms

Strings

- Character encoding (ASCII, Unicode)
- Escape sequences
- String operations complexity
- Mutable vs immutable strings
- String manipulation algorithms

Linked Lists

- Singly linked lists
- Doubly linked lists
- Circular linked lists
- Operations complexity
- Implementation
- Applications

Stacks & Queues

- Stack implementation
- Queue implementation
- Operations complexity
- Applications
- Monotonic stack/queue
- Min stack implementation

Hash Tables

- Hash functions
- Collision resolution
- Load factor
- Rehashing
- Open vs closed hashing
- Complexity analysis
- Applications

Recursion

- Principles of recursion
- Advantages/disadvantages
- Tail recursion
- Mutual recursion
- Recursion vs iteration
- Stack overflow/underflow

Sorting Algorithms

- Bubble sort
- Selection sort
- Insertion sort
- Quick sort
- Merge sort
- Complexity analysis
- Comparison-based vs non-comparison-based
- When to use which algorithm

Searching Algorithms

- Linear search
- Binary search
- Complexity analysis

Trees

- Binary trees
- Binary search trees
- Tree traversal
- Height calculation
- Validation
- Applications

Practical DSA Problems

- Fibonacci with recursion
- Array sum with recursion
- Prime check with recursion
- Binary search with recursion
- Reverse string with recursion
- Palindrome check
- Anagram check
- Find uncommon elements between arrays
- Remove duplicates from linked list
- Detect cycle in linked list
- Reverse linked list
- Find nth node from end
- Sort stack using temporary stack
- Reverse queue using recursion

- Implement queue using stacks
- Balance parentheses
- Find least/most frequent elements
- Two sum problem
- Maximum subarray sum
- Longest substring without repeating chars
- Implement hash table
- Number of islands problem
- Generate all combinations
- Find missing number
- Find median of two sorted arrays
- Implement LRU cache
- Dijkstra's algorithm
- Tarjan's algorithm
- Backtracking problems
- Segment tree implementation
- Sparse array handling

■ Weekly Learning Schedule

I recommend focusing on one category per week:

****Week 1-3:**** JavaScript (deep dive)

****Week 4-5:**** Node.js & Express

****Week 6:**** MongoDB

****Week 7-8:**** React.js

****Week 9-11:**** Data Structures & Algorithms

Each day, mix theory with practical implementation. Spend:

- 60% on practical coding exercises
- 40% on understanding concepts